Towards Automatic Engagement Recognition of Autistic Children in a Machine Learning Approach

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***Abstract*—Children suffering from autism have been continue rising globally and the early intervention may significantly improve the social and cognitive capacity of them. The autism treatment in China is still largely based on training courses and interventions, nevertheless, lacking of professional teachers is a severe problem. Traditional individual training classes are mainly based on teachers' evaluation of students' emotional feedback, and there remain prejudices. The continuous information technology advance, such as ML, have been inspiring novel diagnosis and treatment for autistic children. Note that the majority of the current research on emotional recognition of autistic children have not been conducted on the autistic children in China, in this paper, we aim to conduct a emotional recognition study for them. We study the one-to-one autism training class, and propose an automatic engagement recognition framework to track the behaviors and affection of autistic children by ML. With the improved spatial temporal graph convolutional neural network, our experiments based on the available datasets show that the behavior recognition accuracy reaches beyond 90% and the affection recognition achieves 99%. The result demonstrates that the proposed system can analyze the relationship among arousal, valence and expression category for the autism training class.**

Keywords—autism, action recognition, facial expression recognition

I. INTRODUCTION

Autism Spectrum Disorders (ASDs) are a category of Pervasive Developmental Disorders (PDDs) that are often described as a kind of disease caused by genetic or other factors which can show social communication disorders and repeated sensory movements at a early stage. As a spectrum disorder, the autism spectrum ranges from mild to severe, and sometimes it takes a lifetime to prove that a patient has the disease [1] [2]. Although autism is a physiological disease, the current major treatment options are based on education and behavioral interventions, drugs are just important supplementary means. But the cost of health care, school and care-givers brought by education and behavioral interventions is very high and often imposes a huge burden on the patient's family [1]. At present, the incidence of ASDs in the world is increasing year by year [3]. According to the investigation of the diagnosis rate of children with autism by Deborah L. Christensen et al. in 2012, the diagnosis rate of children with autism in the United States is 1.56% [4]. Although there is no official data in China currently, the article of Yumei Wan in 2013 estimated that the prevalence of ASDs is 24.5/10000, which is considered that there may be 0.25% of children with autism in China [5], related research is urgently needed.

At the same time, Matson et al.'s research shows that the sooner autism patients are discovered, the better the effect of intervention treatment will be [6]. Interventions in childhood autism treatment are the most effective, it is of great significance to pay attention to the treatment of this special group of children with autism. In the process of analyzing children with autism, Rudovic et al. of MIT Media Lab found that children with autistic backgrounds of different cultures had different performances under the same external stimulus, however, due to the small number of experimental samples and the large differences in experimental individuals, they are unable to make causal explanations for behavioral differences in different cultural backgrounds [7]. At the same time, Chinese autistic children are not included in most of the foreign research institutes, so there are still many gaps in this area to explore.

We intend to carry out emotional recognition research for autistic children in China, and combine the related technologies such as DL to realize personalized recognition that suits the characteristics of Chinese children's performance. It is conducive to effectively adjust the feedback of autistic children on the teaching content of individual training classes, so as to realize the real "personalized learning teaching", which will be helpful for autistic teaching to step towards informationization and intellectualization.

II. PROBLEM STATEMENT

At present, researches on autism start late in China, treatments of autistic children are mainly based on educational intervention, supplemented by drug therapy [8]. Teachers and doctors are more likely to guide the performance of autistic children. Studies have shown that the rehabilitation of autistic children in special education schools has achieved certain results, but those schools have a small number of teachers and fewer high-education teachers [9]. Under this situation, how to assist teachers and ease the burden has become a researchable

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direction, teachers need to observe and analyze the behavior of children with autism, which provides space for the application of computer technology. Therefore, a researchable direction in the teaching field of autistic children is combining computer technology with traditional diagnosis and treatment methods, using computer technology to assist teachers in observing and analyzing the behavior of autistic children, reducing the burden on teachers and enabling teachers to accumulate experience quickly to help more children with autism.

Previously, there have been many studies abroad that combine artificial intelligence with interventional therapy for autism [10] [11]. One of the important directions is to quantify the external performance of autistic children [7]. In 2018, this quantitative method was applied to the teaching scene of autistic children with robot participation. A deep learning model was proposed to analyze the behavioral characteristic of autistic children, hoping to help robots understand children's behaviors and emotions [12]. In the same year, Qandeel Tariq et al. proposed to apply machine learning algorithms to help doctors shorten the diagnosis time of autistic children [13].

Thus, there are much data in the world to support the promotion of autism education. Different races, cultures and other factors have a certain impact on the establishment of autism model, so it is particularly important to establish a database based on Chinese children. However, due to the problems of data privacy and reliability of information technology, parents and relevant organizations in China are skeptical about the feasibility of this scheme, which makes it difficult to establish a database. This project aims to design a gesture and expression recognition scoring system for autistic children through the original database to assist teachers in autism classroom, which can also facilitate the establishment of the database for autistic children in China.

III. Algorithm Design

A. Basic Idea

Taking the video as the carrier and the classroom teaching as the background, the design of students' emotional perception scheme is put forward to assist teachers to grasp the student's emotional performance during the classroom teaching and present the student's situation during class with data. The design of the program relies on the actual need of the cooperative autism rehabilitation education institution, and the teaching scenario, task, and feedback are modeled according to real cases. The scenario is a one-to-one teaching session in the actual classroom, that is, a individual training session. Every teaching task is about 3 minutes and the video duration depends on the single teaching task time, which starts with the extraction of the current teaching task and ends up with the student completing the teaching task or timeout. The camera takes an unobstructed fixed-angle view of the sample, the data sample obtained are a series of short videos and teaching plans corresponding to the classroom teaching content. The basic idea of the research based on individual training class is shown in Fig. 1.

In a typical one-to-one ABA training class for autistic children, several roles are involved including teachers, autistic students, parents, and assistants. The content arrangement of each classroom session is selected from the activity list by the teacher, that is to say, the task of the classroom activity is limited and definite, so does the feedback action. The list of classroom activities is shown in Table I.

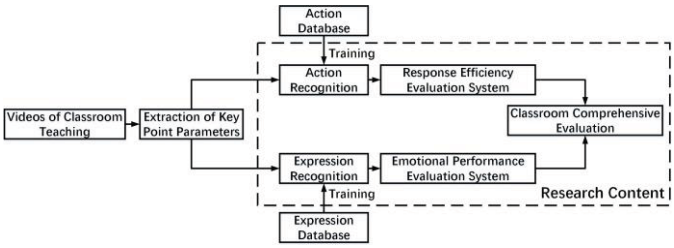


Fig. 1 Research Content

TABLE I. List Of Classroom Activities

Classroom Category	Classroom Activities
Imitation	One-step Action Imitation
	Two-step Action Imitation
	Simple Finger Rumor Imitation
	Moving Cup Imitation
	.....
Perception	Matching Of Physical Objects With Physical Objects
	Matching Physical Objects With Cards
	Matching Cards With Cards
	Visual Conversion
	.....
Cognition	Saying Simple Solutions To Problems
	Finding Obvious Differences Between The Two Objects
	Fruit Card Recognition
	Naming Of Fruit Cards
	Naming A Variety Of Reasons For A Result
	.....
Language Comprehension And Expression	One-step Action Instruction
	Two-step Action Instruction
	Three-step Action Instruction
	One-step Action Continuous Instruction
	Action Instructions With Associative Words
	Sorting Of Three Cards
	.....
Social And Emotional	Answer By Name
	Identifying And Naming Familiar People
	Identifying The Belongings Of Oneself And Others

B. Design of Emotional Evaluation Index

Emotion is one of the external manifestation of students in classroom activities, which is a direct feedback to the current teaching tasks, it has its own value to study students' emotions in classroom learning activities. Therefore, it is particularly important to pay attention to the emotional change of autistic children in class. At the same time, emotion can be reflected by the student's classroom participation and expression information. Thus, we pick them as indicators for relevant research.

1) Emotional performance score based on participation: Children with autism have the most obvious behavior during the interaction process, and many of the literature focus on the definition of participation represented by actions. The object of this project is the actions and expressions of autistic children. Therefore, we choose to define participation from the perspective of action referring to Kim's definition in 2008 [14], which is shown in TABLE II, that is, responsibility of autistic students to current teaching tasks. Kim divides it into 0-6 grades according to the therapist's instruction repetition degree for autistic children and the children's responses. Each grade gives different scores, and the core logic is that the fewer instruction repetitions, the higher participation. Considering the actual situation of the project, the picture captured by the camera does not include the teacher's actions. In addition, since the teacher's instructions are often expressed in the form of voice, but the

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current data acquisition scheme does not include the recording function, so the logic is redesigned. Considering the actual one-to-one teaching scenario, the teacher will assign teaching tasks to the student, and the student need to respond correctly. Based on the correct response, the student's completion degree and completion time are available. Setting the action completion time as t and the duration of a single teaching session as m, the E = t/m can be considered as the student's participation degree in the current teaching task. The smaller the E, the faster the response of the student under the same teaching task, that is, the higher the participation of the student, the higher the E, which indicates the student's response slower, when E = 1, that is the student fails to respond correctly during the task time, and the participation is the lowest.

TABLE II. KIM'S 0-5 RICHTER PARTICIPATION CODING

Participation Level	Meaning	Example
5	High Participation	Immediately responding to the content
4	Medium Participation	Needing the guidance of Teachers
3	Low Participation	Teachers are required to repeat 2-3 times
2	Participation Is Extremely Low	Teachers are required to repeat more than three times
1	Non-participation	No response
0	Escaping	Going away

2) Emotional performance score based on expression:

There are many ways of human emotional expression, physiological signals such as blood pressure, heart rate, hormone levels and so on, external manifestations such as rich facial expressions, tone, and limb movements, etc. Taking the privacy issues and the difficulty of data acquisition in to account, it is clear that choosing the external manifestation is better. In the collected video signals, we can get most of the emotional characteristic from facial expressions, so facial expressions can be used as emotional expressions of autistic students. There are two types of facial expression recognition in the academic circles. The first one is the discrete model which recognizes facial expressions as happy, sad, angry, surprised, nervous, etc [15], the other is the continuous model, such as the PAD (pleasure, arousal, dominance) model proposed by Mehrabian in 1996 [16], the EAP (Evaluation, Activity, Power) model proposed by Cowie in 2001 [17], and Arousal-Valence two-dimensional continuous model proposed by Russel in 1980 [18], etc. For continuous models, the expression can be scored directly from its dimension because of the determination of its continuous value. Taking the Arousal-Valence model as a example, its two dimensions represent the degree of pleasure and intensity of emotion, and the score is between [-1, +1], for the discrete model, when the more people are involved and the higher degree of the satisfaction, the more positive the feeling and the higher the emotional intensity [19], so it can be scored according to the type of emotions. This experiment chooses to identify expressions based on discrete models.

C. Evaluation Algorithm

The algorithm consists of two parts, one based on action-based engagement scores and one based on expression-based Arousal and Valence scores. The two parts also need to acquire data and extract feature points firstly, we use video information (the classroom camera recording) and corresponding teaching task information (teaching activity plan) as the data to be processed, then feature points are extracted from video using the OpenPose, a open source framework. The feature points extracted are divided into two parts for action recognition and expression recognition, and the scores are based on the recognition results.

The main research of this paper is action and expression recognition based on autistic children's classroom situation. So the work is mainly the model and algorithm design based on the two parts.

The main purpose of action recognition is to recognize the action matrix and score the degree of participation. Based on the information obtained from the teaching plan, we generate the correct action label, extract the feature points from the actual classroom video and transform them into the feature points matrix, so as to predict the action of autistic children, if the prediction result is different from the label, the participation degree is set to the lowest directly. If they are consistent, the action completion degree of the corresponding column of the matrix is calculated according to the label of the core feature points of the correct action. Since the matrix column label is the frame label, and the OpenPose converts video to feature point data in the frame unit, the frame selection is uniformly distributed according to the time, so the frame number naturally indicates the time information. When an action is disassembled, there must be angular and distance information between its core feature points. This information can be used to measure whether the action is started or completed. The frame number t of the current picture is obtained. If the total number of frames is m, the participation score can be set to E = t/m.

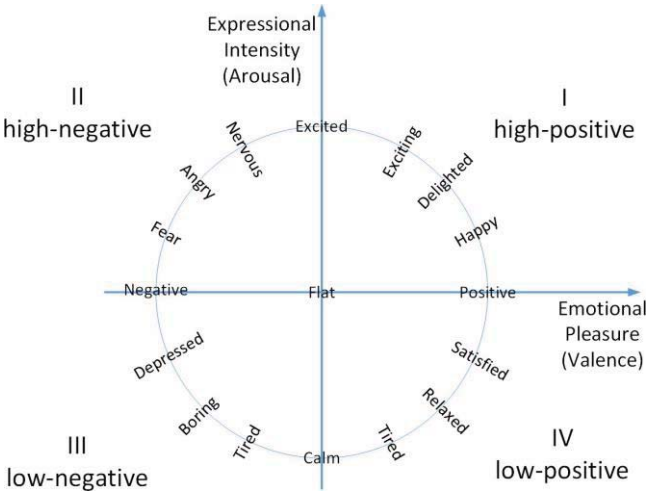


Fig. 2 AN AROUSAL-VALENCE TWO DIMENSIONAL CONTINUOUS MODEL [18]

The algorithm of graph convolutional neural network is used for action recognition. Considering that the duration of the teaching of students in the one-to-one teaching class of autism is generally no more than 3 minutes and the actions that students should complete are simple and clear, the teaching situation in the video can be completely converted into a matrix of actions, and the form is referencing to the ST-GCN network model proposed by the Chinese University of Hong Kong in 2018 [20]. The model expands the graph convolutional neural network to

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the spatiotemporal graph convolutional neural network, and adds a spatial structure in the form of frame to in simple graph data. The spatial feature of the action is represented by the change of the joint point of the same frame, and the temporal feature during the motion is represented by the change of the same joint point between different frames. The method to score the action participation is calculating the start time of the action. When the action starts, the coordinates of the feature point will change. Considering the connection of the same key point corresponding to the two frames as a vector, the change of the motion can be judged by the vector displacement and the angle change, then the start time of the action can be obtained, thereby the participation score can be calculated. Euclidean distance (Eq. 1) and vector cosine (Eq. 2) are commonly used to represent the difference between vectors. The former indicates the difference in displacement between vectors, which can reflect the range of the change in action, the greater the Euclidean distance, the greater the range of action change. The latter indicates the direction difference between the vectors, when the value is closer to 1, the closer the direction change of vectors is to 0◦, and the value is closer to 0, the closer the direction change of vectors is to 90◦.

$$d_{12} = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \quad (1)$$

$$\cos(\theta) = \frac{x_1 x_2 + y_1 y_2}{\sqrt{x_1^2 + y_1^2}\sqrt{x_2^2 + y_2^2}} \quad (2)$$

For the expression recognition part, the aim is to score the expression matrix with Arousal-Valence [18]. The expression feature point information output by feature point extraction module is the same as the action feature point information output, which is a matrix with row label as frame sequence and column label as feature point coordinates. In the expression analysis module, because the Arousal-Valence model is a continuous model, if the expression analysis is done directly, it will become a regression problem. The expression intensity parameter A and the expression pleasure parameter V can be obtained by directly inputting the matrix. If the expression recognition is regarded as a discrete model, it is a classification problem. The specific expression is obtained and then mapped.

The algorithm used for expression recognition is Random Forest. Random Forest is a common method of Ensemble Learning [21] in machine learning. The core idea of it is to construct multiple decision trees and then vote on the classification result to produce the final result. Sampling with replacement and sampling randomly of the features when generating the decision tree, which cause the correlation between the decision tree is very low, therefore it can effectively reduce over- fitting, and the accuracy of the final classification result is also improved. At the same time, the decision tree can also be used to score the importance of the feature, the higher the score, the greater the contribution of the feature to the result. The expression recognition module directly outputs the label of expression when the sequence of feature points based on pictures is input, that is to say, uniform sampling for video tags is needed. According to the definitions of Arousal and Valence, correlate the scores of the two dimensions with the expression tags. By maintaining the score association table, the Arousal and the Valence scores of the expression can be obtained by looking up it.

IV. Experiment And Result

A. Dataset

In the experiment, the data of autistic children have not been obtained for some objective reasons, so the public data set is used. First, from a medical point of view, even if the movement and the expression of autistic children are different from ordinary people, from the computer point of view, the composition of the movement can be characterized by the positional displacement of the joint, and the composition of the expression may be characterized by the shape change and positional displacement of the facial features. That is to say, both autistic children and ordinary people belong to human beings in the field of actions and expressions, so the feature item of the sample are the same, but the feature value will be different, therefore it does not make a fundamental difference between algorithms. Secondly, the main difference between children and youth lies in their different degrees of physical development, but the input of this project algorithm is feature point data, action feature points are bound to the bone joints, and facial features are bound to the facial features and cheeks, which can minimize the physical gap between the youth and the actual sample children in open data sets.

The action data set uses NTU RGB+D [22], it is based on the single action recognition taken in the laboratory with the fixed camera, which contains 60 actions, 40 of them are daily actions, 9 of them are medical actions, and 11 require col- laboration of more than one person. The scene of NTU RGB+D is laboratory. In the video, the same action is repeated by several people and shot from different angles. Connecting with the one-to-one teaching environment of autistic children, NTU RGB+D has some similarities in fixed scenes and single action. Although autistic children and normal children must have different actions, once the action is converted into joint coordinates, this difference can be reduced. Excluding the difference of action, NTURGB+D and the one-to-one teaching scenario of the partner have good consistency. According to the setting task of the partner's individual training classroom teaching session, this project selected 12 actions from NTU RGB+D that may appear in the classroom interaction as the alternative data for the experiment, they are listed in TABLE III. For instance, action mimicry and clapping are common movements in the classroom, pointing to something belongs to the perceptual task category that is adapted to any pairing, or the cognitive task category such as finding things, standing up can be language understanding and the expression task category adapted to one-step action instructions, etc.

The expression data set is based on NVIE (Natural Visible and Infrared Facial Expression), which divides all the pictures into six emotions: happiness, surprise, fear, sadness, disgust and anger. Some pictures contain emotional intensity (Arousal) and emotional pleasure (Valence) labels, and the value is between $[-1, 1]$.

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TABLE III. EXAMPLES OF SELECTED ACTIONS

Cheering	Drinking	Waving	Reading	Shaking Head	Writing
Picking Up	Sitting Down	Standing Up	Clapping	Tearing Off	Pointing Out Something

B. *Engagement based Action Recognition*

1) Establishment of Action Recognition Model: As shown above, the model used in action recognition is ST-GCN and the database is NTURGB+D, when the data are input into OpenPose, the BODY_25 model is used to extract feature parameters. The data obtained are a series of JSON files, including the ordinates and the confidence of each point. The possible error of OpenPose and the corresponding solution are as follows: 1) The number of people detected is not the same as the actual, specifically, there is only one person in the video, but two sets of coordinates are detected. Because OpenPose does not search for people one by one, it is possible to detect a part as the human body's, we choose to calculate the number of zero value in the second set of coordinates, if more than half of the total, it shows that a small part of the background detection is wrong, and the second set is discarded, the first one is chosen. Group coordinates, if the number of zero values is less than the total, it means that it is impossible to judge which group of coordinates is the correct value, both are discarded at the same time; 2) only one set of *coordinates*, but if the number of zero value of coordinates exceeds half of the total, it is discarded; 3) the corresponding coordinate value is set to zero if the confidence level is zero.

The coordinate of feature points in the video are obtained by arranging a set of frame data according to the order of the frame. After processing the video in TABLE III, a total of 11372 coordinate sequences are obtained. The number of the frame varies from dozens to hundreds depending on the length of the video. For convolution, it is necessary to unify the number of frames of the video, normalize all the frame to 260, and make up zero value at the end of whose number of the frame is less than 260. Considering the integrity of the expression change, if the number of the frame is over 260, we select from them according to uniform distribution in existing frames. Because OpenPose also outputs the confidence parameters, the input channels are x, y and c. It is believed that the feature point with high confidence level have greater influence on results.

Because the data set is shot from a fixed perspective and a fixed distance, the position of the target in the video is fixed, and there is no need to search for the size of the person. Video resolution is 1980*1080. Normalized coordinates are obtained by dividing the coordinate by the resolution directly, then the value subtract 0.5 for centralization, and the coordinate are converted to values between (-1, 1). The final matrix is (3, 25, 25), the 3 represents the x dimension, the y dimension and the confidence level dimension of coordinates, the 25 represents 25 feature points.

Totally, 11376 groups of videos are divided into train set, validation set and test set according to the ratio of 6:2:2, when the train set is input, we set the learning rate as 0.01, and reduce it to 10% every 10 times. The batch_size is set to 48, the epoch is set to 70, we choose the SGD optimization algorithm, and the momentu is set to 0.9. The results are shown in Fig. 3.

As we can see from the graph, the loss curve can converge quickly. And when the epoch is 60, the accuracy is 90.9%. It shows that the space-time graph convolution neural network can satisfy the demand of current project using existing data sets.

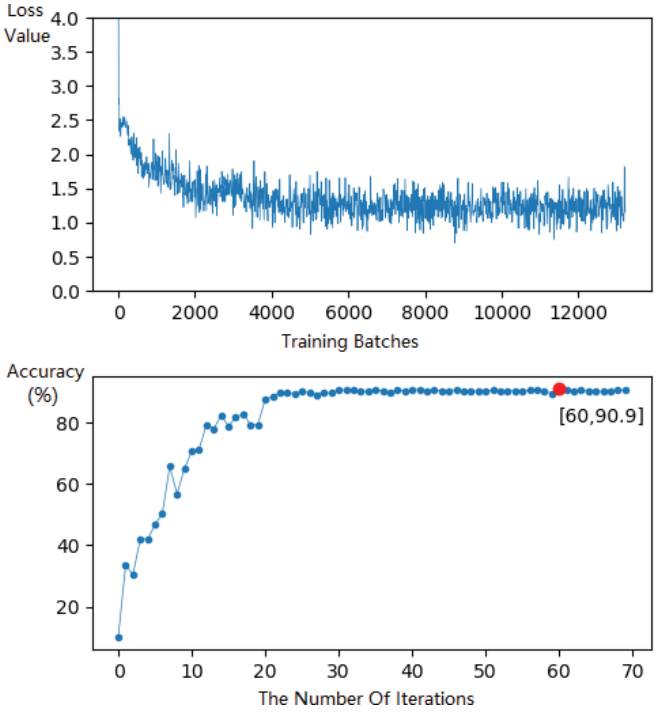


Fig. 3 THE PERFORMANCE OF THE LOSS AND ACCURACY IN THE ITERATION PROCESS

2) Engagement: When the action recognition is accurate, through the ST-GCN, the classification label L of action can be obtained. Looking up the key point sequence v_i $(i \in [1, V])$ corresponding to label L from the database, calculating the displacement of the same key point between different frames for $t \in [1, T-1]$can be regarded as a vector, and the displacement change matrix can be synthesized. The Euclidean distance and the cosine angle are used for calculating the difference, and the obtained feature map is as shown in Fig. 4.

The abscissa of the three groups of pictures is the number of the frame in the video, and the ordinate separately from the top to the bottom are 1) The X-frame and Y-frame figure of 25 feature points. Because there are too many broken lines (50 in total), the legend distinction is not considered. 2) Euclidean Distance-Frame Number Graph of each feature point between adjacent frames. 3) Cosine-Frame Number Graph of each feature point between adjacent frames. Among them, the first group of original video action is "pointing out the object", the second is "cheering", and the third is "waving hands", as we show in Fig. 5. The most obvious key points in the three groups of videos are hands, elbows and shoulders, so the coordinates of feature points with serial numbers of [2, 3, 4, 5, 6, 7] are selected for calculation. As we can see from the graph, when a person readies to move, his limbs will change greatly in a short time, we can find the moment when the action begins referring to this. But it is noteworthy that it is not enough to judge the beginning of a action only from the change of distance or

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direction. In other words, among the action performance of autistic children, there may be a initial action which is changing in distance but unchanged in direction (The movement like raising hands, the distance changes obviously, but the direction is almost unchanged.), as well as a change

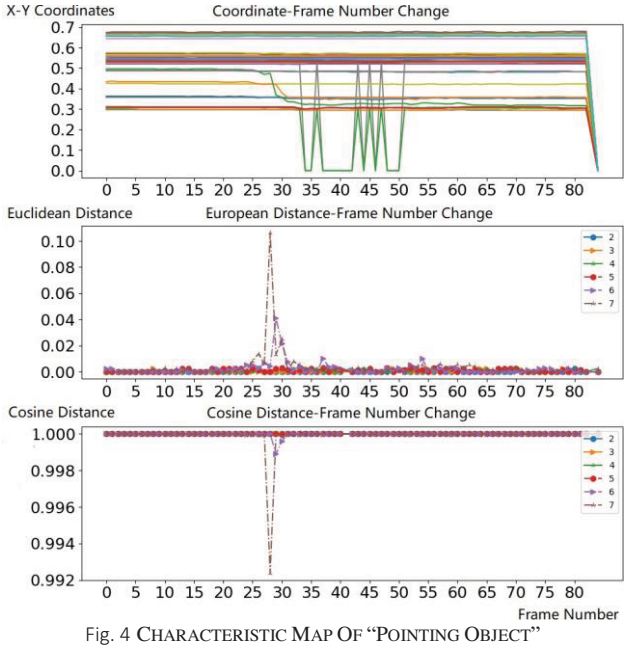


Fig. 4 CHARACTERISTIC MAP OF "POINTING OBJECT"

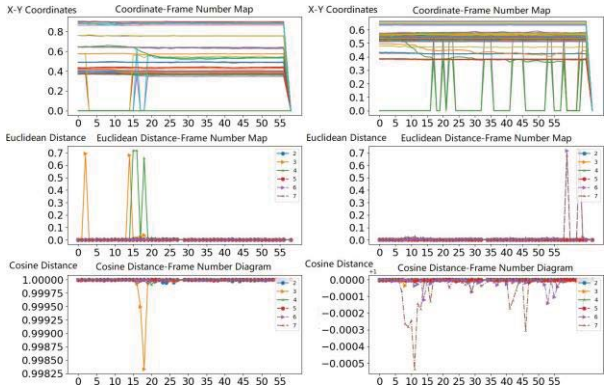


Fig. 5 THE CHARACTERISTIC MAPS OF "CHEERING" AND "WAVING"

only in direction (The hand sways left and right at the beginning of the action. In fact, during the action, the change of distance is much smaller than that of the direction.). That is to say, the initial frame detected by the European distance and cosine angle will be different because of the different action characteristics. But for action, as long as we detect obvious movements, it means that the student begins to give a response. Therefore, this project calculates the Euclidean distance and cosine angle separately, and we choose the minimum value of the initial frame. For individual training classes that expect explicit action feedback, this calculable change can reflect the student's response speed to task instructions and the degree of the student's investment to the current class. By calculating the maximum Euclidean distance or absolute cosine value in two consecutive frames, we can roughly find the specific frame t when completing the action, and then calculate the participation score using $E = t/T$, and the specific algorithm is shown in Algorithm 1.

Algorithm 1: Action Start Frame Evaluation and Rating

Data: Feature extract:F,Model:M,Coordinates Matrix:X,DataBase:DB
Result: Action score: s

$\hat{X} = F(X)$
$label = M\left(\hat{X}\right)$
$key\ points = DB\,(label)$
while *i in key_points* **do**
  for *j in X.shape(0)* **do**
    for *t in X.shape(1)* **do**
      $(E, t)\ .append$
      $((Euclidean_distance\,(X\,[j,t]\,, X\,[j,t+1]))\,, t)$
      $(\Theta, t)\ .append$
      $((\cos\,(X\,[j,t]\,, X\,[j,t+1]))\,, t)$
    end
    $T_E.append\,(t \leftarrow Max\,(|E|))$
    $T_\Theta.append\,(t \leftarrow Min\,(|\Theta|))$
  end
  $frame_begin = Min(T_E \cup T_\Theta)$
  $s = frame_begin/X.shape(1)$
end

C. Arousal and Valence based Expression Recognition

1) Establishment of Expression Recognition Model: Pictures in NVIE are taken continuously, which can be regarded as a series of frames of a video. The NVIE also provides error frames and error mode information. There are 103637 available pictures except those with zero coordinates of feature points.

Due to the difference of the number of video frames, a suitable frame size is selected according to the median and the mode. Each frame coordinate is spliced into a matrix with X label as frame and Y label as feature points. Finally, 1767 matrices, namely video units, are obtained.

Because of the different shooting angles of pictures in NVIE and the different face size of the target, the normalization method can not be dividing the resolution by coordinates as NTURGB+D. The common normalization methods are min-max normalization and Z-score normalization. Considering the advantage of Z-score in distance measure similarity, choosing it for normalization.

For too few video units and the complexity of sequence features based on video units, using picture as a unit, for 103637 samples, the results of several common linear and nonlinear classifiers are displayed in TABLE. IV.

TABLE IV. EXPRESSION RECOGNITION BASED ON THE PICTURE AS A UNIT

Machine Learning Algorithms	Logical Regression Classification	Ridge Regression Classification
Accuracy%	0.335	0.026
Machine Learning Algorithms	Random Forest Classification	Decision Tree Classification
Accuracy%	0.990	0.897
Machine Learning Algorithms	Gradient Descent Classification	Supporting Vector Machine Classification
Accuracy%	0.225	0.764

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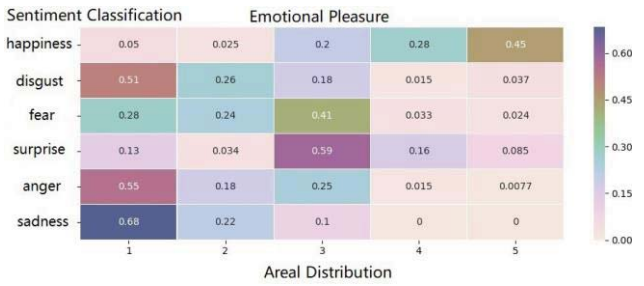


Fig. 6 AROUSAL DISTRIBUTION DENSITY MAP

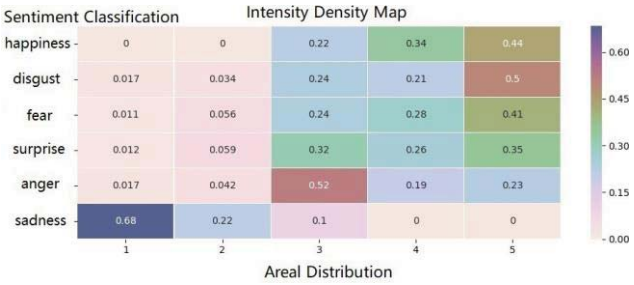


Fig. 7 VALENCE DISTRIBUTION DENSITY MAP

As can be seen from the tab le, for the figure input, nonlinear classifiers perform better than the linear classifier, among them, the random forest algorithm works best because of its playback sampling and the random sampling of feature points which can prevent overfitting. Using the grid tuning to adjust the random forest, the final accuracy is 0.990062 which meets the requirement.

2) Arousal and Valence: We use NVIE for expression recognition training. There are 877 sequences in the NVIE database contain Arousal and Valence tags, and the others can be summarized according to the point coverage and the similarity. We convert label distribution into density map, as shown in Fig. 6 and Fig. 7, the 0-5 area indicates that the score is from low to high, and according to its probability density partition, the score of emotional intensity and pleasure under different labels can be compared. Each dimension is roughly ranked from large to small according to the corresponding values of the area of the sample concentrate: 1) emotional arousal: happiness, surprise, disgust, fear, anger, sadness; 2) emotional valence: happiness, surprise, fear, disgust, anger, sadness. The higher the emotional activity, the more pleasant the performance is, and the higher the student's participation, so the simple mapping is carried out in order, as shown in TABLE V.

TABLE V. DISCRETE EMOTION CLASSIFICATION

Emotional Classification	Happiness	Disgust	Fear	Surprise	Anger	Sadness
Emotional Intensity Rating	6	5	4	3	2	1
Emotional Pleasure Rating	6	3	4	5	2	1

However, this kind of scoring method has a serious dependence on data. In the actual system, it is necessary to wait for the sufficient data to verify and revise the existing scoring method according to the professional teachers' evaluations of the student's emotion. If Fig. 8 is based on the real data, from the perspective of the autistic student, it can be explained that when students show the positive emotion to the stimulation of the outside world in the classroom, the students' state is good, the calmer the student's performance is, and even negative emotions are generated, indicating that the student's state is worse.

V. CONCLUSION

The performance evaluation of our proposed system is divided into two categories with three aspects. The first category is the participation score based on the student's response speed to the classroom instruction. We define the scoring method based on the motion start frame judgment (start frame/total frame number). At the same time, considering the diversity of the definition and quantification of participation in academia, the emotional indicator is also used as an auxiliary parameter. The second category is the emotional score based on the student's emotional expression of the classroom atmosphere, namely the activity score and the pleasure score based on the Arousal-Valence model. Our experiments based on the available datasets demonstrate the technical feasibility to automatically recognize the engagement of autistic children with high accuracy. In the future, we plan to construct an emotional representation of the autistic students' training class based on these three parameters.

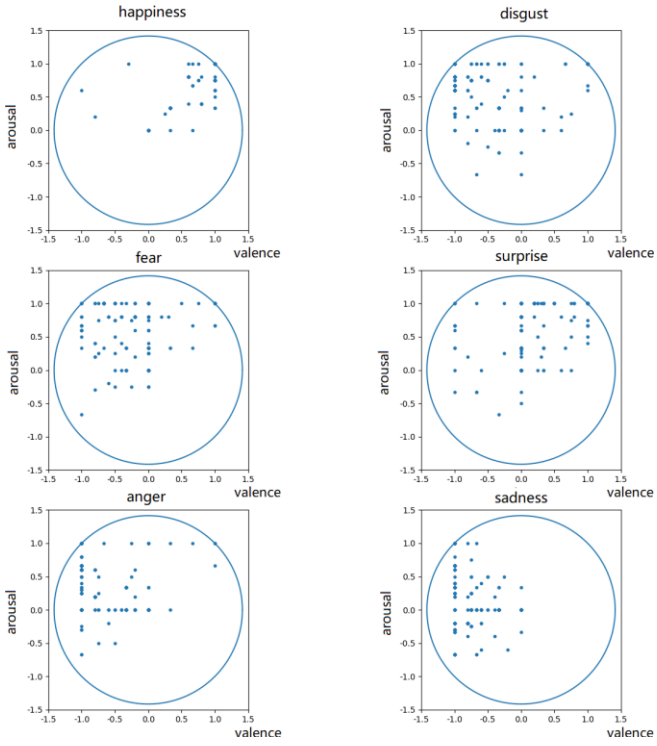


Fig. 8 CHARACTERISTIC MAP OF "CHEERING" AND "WAVING HAND"

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For implementation, considering the actual demand, this work is based on OpenPose which is easy to use and has many versions. At the same time, it contains the expression feature point and the action feature point extraction modules. The action scoring module is divided into two parts, the first part is the motion recognition based on the spatiotemporal graph convolutional neural network, and the second is the initial frame detection and scoring based on the Euclidean distance and cosine value, the accuracy is sufficient for training data. The expression scoring module is also divided into two parts. First of all, the expression tag is obtained by random forest, then the expression label is used to score the mapping relationship between the activity rating and the pleasure rating, and good results have also been achieved.

VI. Future Work

Because of the particularity of the research-oriented group, autistic children need real information in the classroom to test the feasibility in the actual scenario. After we collect the autistic children's real data, future work can be carried out from these aspects:

1) Current research mainly identifies the action of one person, but the real classroom teaching process is a one-to-one interaction process. If teachers' behaviors are included in the scope of recognition, it will be more realistic and more accurate to identify the classroom interaction between teachers and students.

2) After recognizing the emotional performance of autistic children, we can consider quantifying the result and find out the relationship between the result and the score of classroom participation pleasure and so on, which is conducive to future behavioral statistical analysis.

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